TEXORA · LECTURE SHEET

TQM Tools II: Quality Circles, QFD, Taguchi, PM, BPR and Standards

The people-and-design tools of TQM — quality circles that solve problems where they occur, quality function deployment that translates the customer's voice, the Taguchi loss function with a worked example, productive maintenance, performance measurement, re-engineering, and product standards (BSTI, ASTM) with specifications and tolerances.

TE-4108 · QUALITY ASSURANCE AND MANAGEMENT IN TEXTILES · TOPIC 4 · LEVEL 4 TERM 1

1. Quality circles

A **quality circle** (Ishikawa) is a small voluntary group — typically 6–10 people from the same work area, led by their supervisor — that meets regularly to identify, analyse and solve work-related problems using the basic QC tools of Topic 3, presenting solutions to management for approval. Circles work because the people closest to a process hold its diagnostic knowledge; they endure only when management responds to presentations promptly, implements accepted solutions visibly, and never uses the circle as a channel for grievances it intends to ignore. A weaving-section circle attacking weft breaks with a check sheet, Pareto and fishbone (Topics 3 and TE-4107) is the classic textile example: the tools are simple, the knowledge is local, the savings repeat daily.

2. Quality function deployment (QFD)

QFD translates the **voice of the customer** into engineering characteristics through a matrix — the **House of Quality**: customer requirements (WHATs, with importance ratings) as rows; technical characteristics (HOWs) as columns; a relationship matrix (strong/medium/weak) between them; a "roof" of correlations among the HOWs; competitive benchmarking along the right wall; and target values along the bottom. The HOWs are then deployed through successive houses — product → parts → process → production requirements — so a buyer's phrase becomes a machine setting with an audit trail.

BUYER SAYS (WHAT)	IMPORTANCE	TRANSLATED HOW	TARGET
"Shade must match season to season"	5	Lab-to-bulk ΔE control; spectrophotometer checks	ΔE ≤ 0.8 CMC (2:1)

"Fabric must not shrink in washing"	5	Relaxation/finishing route; dimensional stability test	−3 % to +1 % warp and weft
"No harsh handle"	3	Softener type and dose; silicone finish	Panel handle score ≥ 4/5
"Deliveries never late"	4	Production scheduling; buffer stock of grey fabric	98 % on-time shipment

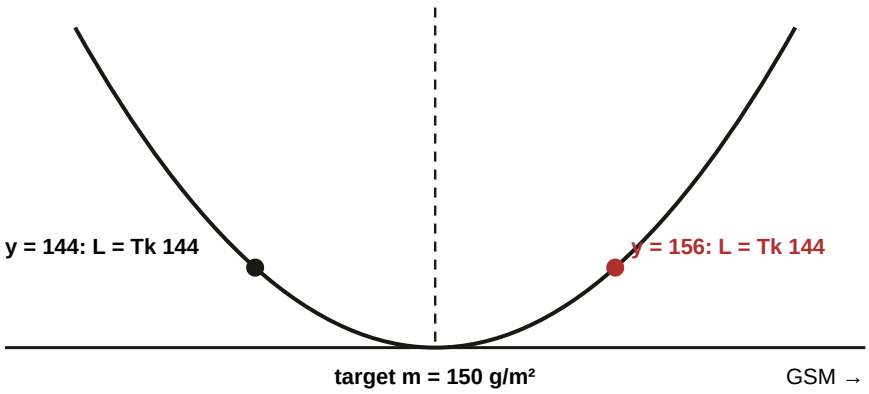
The house's arithmetic (importance × relationship strength, summed per column) tells the mill which HOWs deserve its engineering money first — here shade control and shrinkage control dominate, exactly as a dyeing manager would hope the data to show.

3. The Taguchi loss function

Taguchi's central insight: **quality loss begins the moment a characteristic deviates from its target — not only when it crosses a tolerance limit.** A fabric at the very edge of its GSM tolerance is "acceptable" to the tolerance view and measurably worse to the customer. The loss is modelled as a quadratic:

$$L(y) = k\,(y - m)^2, \quad \text{with} \quad k = A_0 / \Delta_0^2$$

where m is the target, y the actual value, Δ_0 the functional tolerance limit (where the product fails in use) and A_0 the cost incurred at that limit (replacement, claim). For a population, the expected loss is $E[L] = k(\sigma^2 + (\bar{y} - m)^2)$ — variation around the target and off-centre mean both cost money, which is the mathematical form of "reduce variation, centre the process".



Loss grows with the square of deviation — at ±6 from target the loss is already Tk 144 per unit of loss accounting.

Fig. 1 — The quadratic loss function for GSM, target 150 g/m²: zero loss only exactly on target, rising symmetrically — the same value at 144 and 156 g/m².

WORKED EXAMPLE — TAGUCHI LOSS FOR FABRIC GSM

Target GSM m = 150 g/m². The buyer rejects and claims at Δ_0 = 10 g/m² off target, with loss A_0 = Tk 400 per affected roll-unit.
k = A_0 / Δ_0^2 = 400 / 10² = **4 Tk per (g/m²)²**.
A roll at y = 156 g/m² (inside a ±8 specification, and only 6 off target): L = 4 × (156 − 150)² = 4 × 36 = **Tk 144** of loss — despite passing inspection.
A production run with mean $\bar{y}$ = 152 and σ = 3: E[L] = k(σ² + ($\bar{y}$ − m)²) = 4 × (9 + 4) = **Tk 52 per roll**.
Management reading: centring the mean on 150 and halving σ to 1.5 cuts expected loss to 4 × 2.25 = Tk 9 — an 83 % reduction available from process control (Topic 3's charts), invisible to any accounting that counts only out-of-tolerance rolls.

4. Productive maintenance (PM)

Productive maintenance in the TQM context — **total productive maintenance (TPM)** — makes equipment reliability everyone's work: operators perform autonomous care (cleaning, lubrication, tightening, inspection — cleaning as inspection), maintenance specialists handle planned and preventive work, and the organisation tracks **OEE**. TPM attacks the classic losses: breakdowns, set-up losses, minor stops, speed loss, defects and start-up yield loss. Its quality link is direct: a large fraction of "unexplained" variation and defect bursts in textile processes are equipment-condition events in disguise — worn needles, drifting temperatures, dirty sensors. The maintenance record discipline of TE-4106 is TPM's factory floor.

5. Performance measures

TQM measures what it preaches, at process level: defect rates (DPPM/DPMO), first-pass yield, cost of quality (Topic 1), on-time delivery, complaint rate and closure time, audit scores, supplier ratings (Topic 2) — and increasingly balanced views (financial, customer, internal process, learning) so one metric cannot be gamed at the expense of the system. Rules: few measures, defined identically everywhere, plotted over time (Topic 3), reviewed with the people who produce them, and tied to action — a measure nobody acts on becomes wall decoration, then cynicism.

6. Business process re-engineering (BPR)

BPR is the radical counterpart of kaizen: instead of improving an existing process step by step, it asks whether the process should exist in its present form at all, and redesigns it from a clean sheet for dramatic gains in cost, quality, service or speed (Hammer and Champy). Textile example: instead of shaving days off a sample-approval loop of couriers and re-dyeings, re-engineer it around digital shade communication and lab-data sharing — the loop itself changes shape. BPR's risks match its ambition:

failure rates are high, human disruption severe, and it discards accumulated process knowledge; TQM practice uses it sparingly, where incremental improvement demonstrably cannot reach the required performance — and never as a synonym for headcount reduction, which poisons the involvement TQM depends on.

7. Standards awareness: BSTI, ASTM; specifications and tolerances

Product standards convert quality language into testable requirements. The **BSTI** (Bangladesh Standards and Testing Institution) publishes Bangladesh Standards (BDS), many aligned with ISO, and operates product certification relevant to domestic-market textiles. **ASTM International** publishes the test methods and performance specifications widely cited in export contracts for textiles — e.g. the D-series methods for fabric breaking strength, dimensional change, colourfastness evaluations (with AATCC methods), and flammability — alongside ISO and buyer-specific standards. Reading any standard means reading three things precisely: the **specification** (the requirement values), the **test method** (exactly how to measure, conditioning included) and the **tolerance** (permitted deviation, e.g. GSM ±3 %, measurements ±1 cm). Disputes between mill and buyer are almost always located in one of the three: a value measured by a different method, under different conditioning, against a different tolerance. A quality engineer's craft is keeping the three aligned in the contract, the lab and the control plan.

	KAIZEN / CPI	BPR
Change size	Small, continuous	Radical, one-step
Starting point	Existing process	Clean sheet
Participation	Everyone, daily	Project team, top-down
Risk	Low per step	High
Typical horizon	Permanent habit	One redesign episode

Glossary

Quality circle — small voluntary work-area group solving its own problems with QC tools. **QFD / House of Quality** — matrix deployment of customer requirements into technical characteristics and targets. **Loss function** — Taguchi's $L = k(y - m)^2$: cost of deviation from target. **TPM** — total productive maintenance; operator care + planned maintenance; measured by OEE. **BPR** — radical process redesign from a clean sheet. **Specification** — stated requirement values; **tolerance** — permitted deviation from them.

Key points

• Quality circles localise problem-solving; QFD deploys the customer's voice into targets with an audit trail.
• Taguchi: loss starts at any deviation from target, $L = k(y - m)^2$ — worked: k = 4, L(156) = Tk 144, E[L] = Tk 52 at $\sigma = 3$ with a 2-unit bias.
• TPM makes equipment condition a quality variable; performance measures must be few, consistent and acted on.
• BPR is the radical exception, kaizen the rule. Standards (BSTI, ASTM) bind specification + method + tolerance — read all three together.

Practice

1. A characteristic has target 25.0, functional limit $\Delta_0 = 0.5$ with A_0 = Tk 900. Find k and the loss at y = 25.3.
2. A process runs at mean = target with $\sigma = 0.2$ using the k of Q1. Expected loss per unit? What if σ doubles?
3. Why can a quality circle not fix a problem whose cause lies in the purchasing specification? What TQM mechanism handles it instead?
4. A buyer claims GSM is out of tolerance; the mill's lab disagrees. List the three standard elements to align before either side can be right.

Answer key

1. $k = 900 / 0.5^2 = 900 / 0.25 = 3{,}600$. $L(25.3) = 3{,}600 \times 0.3^2 = 3{,}600 \times 0.09$ = Tk 324.
2. $E[L] = k\sigma^2 = 3{,}600 \times 0.04$ = Tk 144 per unit. $\sigma = 0.4$: $E[L] = 3{,}600 \times 0.16$ = Tk 576 — quadrupling: loss scales with the *square* of σ, which is why variation reduction dominates quality economics.
3. A circle's authority and knowledge are local to its work area; a defective purchasing specification is set outside that area, and no amount of local effort compensates a wrong input specification (the team would only learn to cope). The mechanism is cross-functional: quality planning / management review escalates the finding, the specification is corrected with the supplier-partnership process (Topic 2), and the circle's data serve as the evidence. Circles surface such problems; the system must route them upward — one more reason management response speed decides whether circles live.
4. (1) The **specification**: the nominal GSM and whose document states it (contract, buyer standard, BDS/ASTM reference). (2) The **test method**: standard followed, sample size, conditioning atmosphere and moisture state — GSM measured unconditioned after a humid day is a different number. (3) The **tolerance**: the permitted ± % and whether it applies to the roll average or every reading. Until all three are identical on both sides, the two labs are answering different questions.